

Exercise 2: Approximation Theory

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Problem 1 (Weierstrass theorem):

In this exercise, we seek to derive a formal proof of the Weierstrass theorem discussed in the lecture. Recall that the theorem can be stated as follows:

Theorem 1 (Weierstrass). *Given a continuous function $f(x)$ on $a \leq x \leq b$ and an arbitrary positive constant $\epsilon > 0$, it is possible to construct an approximating polynomial $P(x)$ such that*

$$|f(x) - P(x)| \leq \epsilon, \quad \forall a \leq x \leq b \quad (1)$$

Without loss of generality, we assume $a = 0$, $b = 1$ and extend f on the whole $\mathbb{R}$ by setting $f(x) = 0$, $\forall x \notin (0, 1]$, such that f is continuous on $(-\infty, 1]$. (Why can we do this?)

Let $P_n(x)$ be a polynomial of degree $2n$ such that

$$P_n(x) = \frac{1}{J_n} \int_0^1 f(t)[1 - (t - x)^2]^n dt, \quad (2)$$

where $J_n := \int_{-1}^1 (1 - u^2)^n du$ is a constant.

a) Show that

$$f(x) = \frac{1}{J_n} \int_{-1}^1 f(x)(1 - u^2)^n du \quad (3)$$

b) Show that

$$P_n(x) - f(x) = \frac{1}{J_n} \int_{-1}^1 [f(x + u) - f(x)](1 - u^2)^n du \quad (4)$$

where $x \in [0, 1]$. The problem is now to show that this expression approaches zero as $n \rightarrow \infty$.

c) Let $\epsilon > 0$. Use the following facts freely:

- since $f(x)$ is continuous on $[-1, 1]$, there exists a $\delta > 0$ such that $|f(x + u) - f(x)| \leq \frac{\epsilon}{2}$ for each x, u with $|u| < \delta$ and $x, x + u \in [-1, 1]$;
- there exist a positive constant $M > 0$, such that $|f(x)| \leq M$, $\forall x \in [-1, 1]$.

Show that

$$|f(x + u) - f(x)| \leq \frac{\epsilon}{2} + 2M \frac{u^2}{\delta^2}, \quad \forall x \in [-1, 1]. \quad (5)$$

Hint: Think of the case distinction where $|u| \geq \delta$, i.e. $1 \leq \frac{u^2}{\delta^2}$; and where $|u| < \delta$.

d) Using integration by part, show that

$$J'_n := \int_{-1}^1 u^2(1 - u^2)^n du = \frac{J_{n+1}}{2(n+1)}$$

and also show that

$$J_n > J_{n+1}, \quad \forall n \in \mathbb{N}.$$

e) Finally, re-using the answers to the previous sub-problems, prove that

$$|f(x) - P_n(x)| \leq \epsilon, \quad \forall x \in [0, 1] \quad (6)$$

for sufficiently large n .

Problem 2 (Stone-Weierstrass theorem):

The following is a generalization of Weierstrass theorem.

Theorem 2 (Stone-Weierstrass, see Theorem 2.2. in Telgarsky (2021)). *Let function class $\mathcal{F}$ of real-valued functions defined on $[0, 1]^d$ be given as follows:*

- i) Each $f \in \mathcal{F}$ is continuous.*
- ii) For every $\mathbf{x} \in [0, 1]^d$, there exists $f \in \mathcal{F}$ with $f(\mathbf{x}) \neq 0$.*
- iii) For every $\mathbf{x} \neq \mathbf{x}' \in [0, 1]^d$, there exists $f \in \mathcal{F}$ with $f(\mathbf{x}) \neq f(\mathbf{x}')$ ($\mathcal{F}$ separates points).*
- iv) $\mathcal{F}$ is closed under multiplication and vector space operations, i.e. $\mathcal{F}$ is an algebra.*

Then $\mathcal{F}$ is a universal approximator: for every continuous $g : [0, 1]^d \rightarrow \mathbb{R}$ and $\epsilon > 0$, there exists $f \in \mathcal{F}$ with

$$|f(\mathbf{x}) - g(\mathbf{x})| \leq \epsilon, \quad \forall \mathbf{x} \in [0, 1]^d.$$

a) Consider unbounded width networks with one hidden layer:

$$\mathcal{F}_{\sigma, d, m} := \mathcal{F}_{d, m} := \{x \mapsto a^\top \sigma(Wx + b) : a \in \mathbb{R}^m, W \in \mathbb{R}^{m \times d}, b \in \mathbb{R}^m\}. \quad (7)$$

$$\mathcal{F}_{\sigma, d} := \mathcal{F}_d := \bigcup_{m \geq 0} \mathcal{F}_{\sigma, d, m}. \quad (8)$$

Using Theorem 2, show that $\mathcal{F}_{\cos, d}$ is universal, where $\cos : \mathbb{R} \rightarrow \mathbb{R}$ is the cosine function.

b) Show that Theorem 2 does not hold if

- i) condition (i) in Theorem 2 does not hold;
- ii) condition (ii) in Theorem 2 does not hold;
- iii) condition (iii) in Theorem 2 does not hold;
- iv) condition (iv) in Theorem 2 does not hold.

References

Matus Telgarsky. Deep learning theory lecture notes. <https://mjt.cs.illinois.edu/dlt/>, 2021. Version: 2021-10-27 v0.0-e7150f2d (alpha).